DATABASE MANAGEMENT SYSTEMS • PROJECT REVIEW 2

DESIGN AND IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM FOR LIBRARY CIRCULATION AND DIGITAL RESOURCE MANAGEMENT

Focus: Conceptual E-R Modeling, Relational Schema Conversion, Normalization (1NF–3NF), Integrity Constraints & Schema DDL

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Technology: MySQL 8.0 & Python 3 | Milestone: Review 2

Project Evolution: Progress from Review 1 to Review 2

Review 1 Scope & Problem Definition

- **Problem Statement:** Manual registers and isolated spreadsheets cause duplicate data, untracked book copies, lost fines, and zero visibility into queue waiting lists.
- **Objectives Set:** Design an integrated, centralized relational database to manage print books, digital resources, patrons, circulation, and fines.
- **User Roles Identified:** Librarians (daily operations), Admins (policies & reports), and Members (students, faculty, public).
- **Initial Scope:** High-level requirements gathering and initial feature specification.

Review 2 Deliverables Achieved

- **Comprehensive E-R Model:** Identified 15 distinct entities, cardinalities, associative entities, and relationship attributes.
- **3-Tier Catalog Architecture:** Split Book (title work), Edition (ISBN publication), and Copy (physical barcode/shelf item).
- **Relational Mapping:** Transformed conceptual ER model into normalized relational schemas with full foreign key constraints.
- **Formal Normalization Proof:** Verified progression through 1NF, 2NF, and 3NF/BCNF to prevent update, insertion, and deletion anomalies.
- **MySQL 8.0 DDL Scripts:** Implemented 01_schema.sql with primary keys, checks, enums, cascades, and indexes.

System Requirements & Circulation Rules Specification

1. Functional Scope

- **Catalog Control:** Manage books, editions, authors, publishers, and categories.
- **Copy Tracking:** Unique accession numbers per physical copy with real-time status.
- **Circulation:** Issue, return, and loan renewal operations.
- **Hold Queue:** Ordered waitlist reservations for out-of-stock titles.
- **Fines & Audits:** Overdue fee calculation and payment tracking.
- **Digital Assets:** E-books, journals, and patron access duration logging.

2. Tiered Policy Table

- **Student Policy:** Max 3 books concurrently | 14-day loan period | ₹2.00 / late day fine.
- **Faculty Policy:** Max 6 books concurrently | 30-day loan period | ₹1.00 / late day fine.
- **Public Policy:** Max 2 books concurrently | 7-day loan period | ₹5.00 / late day fine.
- **Separation Rationale:** Rules live in `member_policy` table so admins adjust loan rules dynamically without modifying codebase.

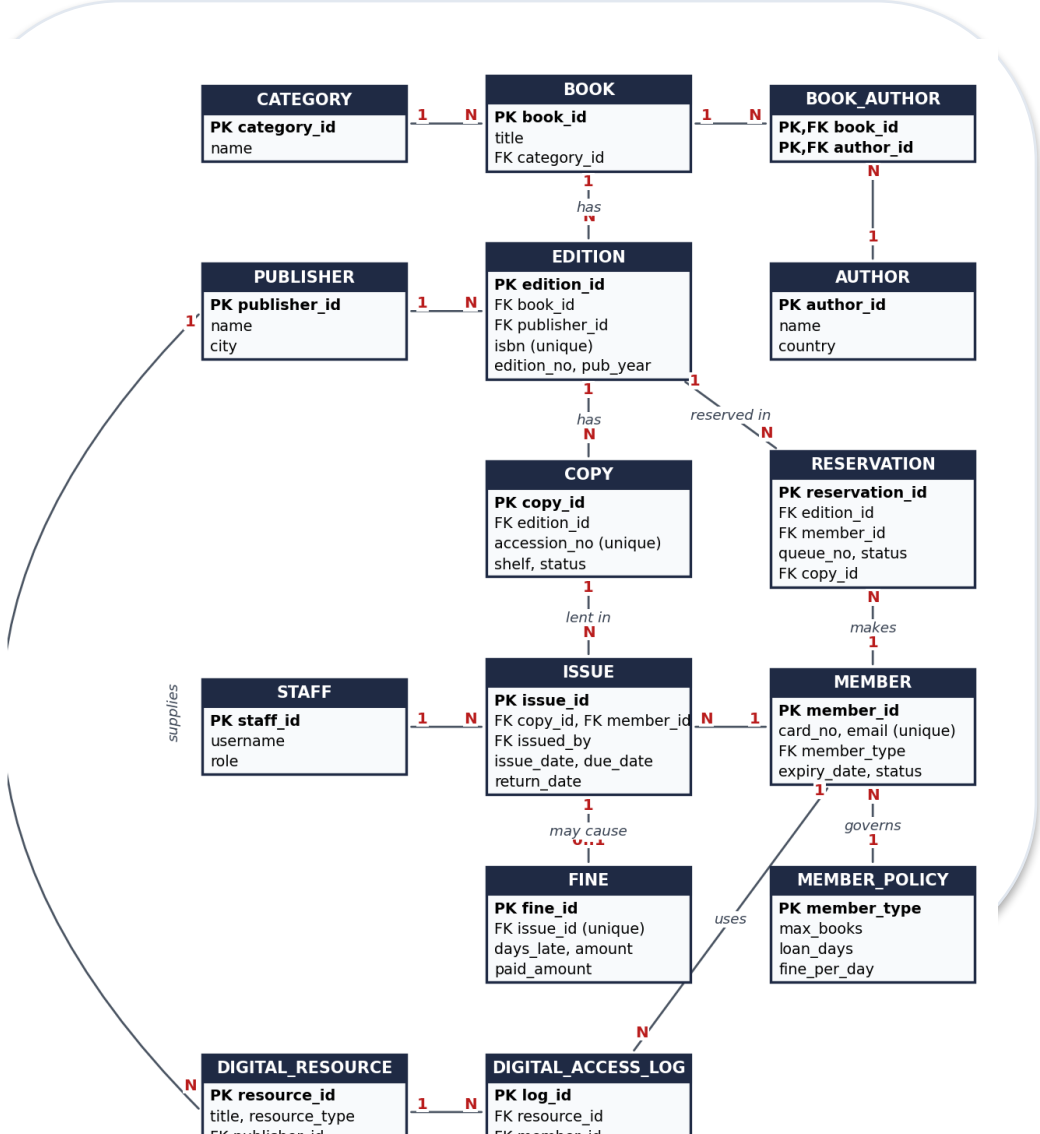
3. Integrity Constraints

- **₹50 Fine Lock Rule:** A patron with > ₹50 in unpaid fines cannot borrow any new books.
- **1 Active Issue / Copy:** A physical copy cannot be issued if status is already ISSUED or ON_HOLD.
- **Reservation Trigger:** A reservation can only be placed when 100% of physical copies are checked out.
- **FIFO Hold Transition:** Returned books automatically assign to next WAITING reservation and shift to READY.

Conceptual Design: Entity-Relationship (ER) Architecture

Core Entities & Relationship Mappings

- **Publisher (1) —< (N) Edition:** A publisher produces many published book editions.
- **Book (N) >—< (M) Author:** Many-to-many relationship resolved using `book_author` bridge table.
- **Book (1) —< (N) Edition:** One master title work has multiple physical published editions.
- **Edition (1) —< (N) Copy:** One edition has multiple physical inventory copies on library shelves.
- **Copy (1) —< (N) Issue:** A copy appears in loan history; only one active issue at a time.
- **Member (1) —< (N) Issue:** Patrons borrow books governed by their linked `member_policy`.
- **Issue (1) —< (0..1) Fine:** Late returned books generate at most one fine record.
- **Edition (1) —< (N) Reservation:** Waitlist entries queue members for checked-out editions.



Key Structural Decision: The 3-Tier Catalog Model

Tier 1: BOOK (The Work)

- **Concept:** Represents the intellectual literary work regardless of publication format.
- **Attributes:** `book_id` (PK), `title`, `category_id` (FK).
- **Example:** 'Database System Concepts'
- **Advantage:** Title name, authors, and classification are defined exactly once. No duplication when multiple editions are purchased.

Tier 2: EDITION (Publication)

- **Concept:** Represents a specific release from a publisher with page count and year.
- **Attributes:** `edition_id` (PK), `book_id` (FK), `publisher_id` (FK), `isbn` (UNIQUE 13-digit), `edition_no`, `pub_year`, `pages`.
- **Example:** 7th Edition, McGraw-Hill, 2019, ISBN: 9780078022159.
- **Advantage:** Strictly separates publication metadata from physical items.

Tier 3: COPY (Inventory)

- **Concept:** Represents a specific physical book sitting on the library shelf.
- **Attributes:** `copy_id` (PK), `edition_id` (FK), `accession_no` (UNIQUE barcode), `shelf`, `status` (AVAILABLE, ISSUED, ON_HOLD, WITHDRAWN).
- **Example:** Barcode `ACC-CS-001` on shelf `CS-A1-01`.
- **Advantage:** Enables accurate per-item circulation history and real-time availability tracking.

Database Normalization: 1NF to 3NF / BCNF Analysis

Step-by-Step Normalization Breakdown

- **First Normal Form (1NF) Compliance:** All attributes contain atomic values. Multi-valued fields (e.g. multiple authors for a book) are eliminated by introducing the associative table `book_author(book_id, author_id)`. ISBNs, phone numbers, and dates are stored in singular standardized columns.

- **Second Normal Form (2NF) Compliance:** All non-prime attributes are fully functionally dependent on the primary key. In composite primary key tables (like `book_author`), no non-key attribute depends on just a subset of the primary key. Author details live in `author`, title details live in `book`.

- **Third Normal Form (3NF) Compliance:** Elimination of transitive functional dependencies ($X \rightarrow Y, Y \rightarrow Z$). Example: A member's borrow limit and daily fine rate depend strictly on `member_type`, not `member_id`. Moving `(member_type, max_books, loan_days, fine_per_day)` into `member_policy` ensures that policy changes do not cause update anomalies across thousands of member rows.

- **Boyce-Codd Normal Form (BCNF):** Every determinant in every relation is a candidate key. All functional dependencies $X \rightarrow Y$ satisfy the condition that X is a superkey.

Relational Schema Specification & Integrity Constraints

Relational Table Catalog (15 Tables)

- **Master Entities:** `publisher`, `author`, `category`, `staff`, `member_policy`
- **Catalog Schema:** `book`, `book_author`, `edition`, `copy`
- **Circulation Schema:** `member`, `issue`, `fine`, `reservation`
- **Digital Resource Schema:** `digital_resource`, `digital_access_log`
- **Primary Key Policy:** Auto-increment synthetic integers for internal joins; natural UNIQUE candidate keys for domain identifiers (`isbn`, `accession_no`, `card_no`).

Enforced Database Constraints

- **Domain Constraints:** CHECK (`max_books` > 0), CHECK (`loan_days` > 0), CHECK (`fine_per_day` >= 0), CHECK (`days_late` >= 0).
- **Enum Constraints:** `role` ENUM('LIBRARIAN', 'ADMIN'); `status` ENUM('AVAILABLE', 'ISSUED', 'ON_HOLD', 'LOST', 'WITHDRAWN'); `member_type` ENUM('STUDENT', 'FACULTY', 'PUBLIC').
- **Referential Integrity:** Foreign key cascades on book deletion (`ON DELETE CASCADE`), while restricting deletion of copies currently out on loan (`RESTRICT`).
- **Uniqueness:** Enforced unique ISBN-13, barcode accession numbers, member card numbers, and staff usernames.

Database Implementation: Schema DDL (`01_schema.sql`)

DDL Implementation Highlights in MySQL 8.0

- **Engine & Character Set:** Configured all tables using MySQL `InnoDB` engine for robust ACID transaction support, foreign key constraints, and crash recovery. Default charset `utf8mb4` with `utf8mb4_unicode_ci` collation.
- **CHECK Constraint Enforcement:** Leveraged MySQL 8.0.16+ native CHECK constraint enforcement to ensure monetary fine amounts, days overdue, and book allocations never accept negative or invalid values.
- **Temporal Tracking:** `issue_date`, `due_date`, `return_date`, and `assessed_on` typed as DATE/DATETIME with DEFAULT `CURRENT_TIMESTAMP` for reliable circulation auditing.
- **Indexing Strategy:** Primary keys indexed automatically; added secondary indexes on `member.card_no`, `copy.accession_no`, `edition.isbn`, and `issue.return_date` to accelerate high-frequency circulation queries.

Summary of Review 2 & Roadmap to Final Review 3

Review 2 Deliverables Complete

- **Conceptual ER Model:** Complete entity identification & relationship cardinality defined.
- **3-Tier Normalization:** Separated Book, Edition, Copy; normalized to 3NF/BCNF.
- **Relational Schema:** Full 15-table relational design with primary and foreign keys.
- **Integrity Architecture:** CHECK constraints, ENUM validations, and uniqueness rules.
- **MySQL DDL Generation:** `01_schema.sql` scripted, verified, and executed in MySQL 8.0.

Target Roadmap for Review 3

- **Data Seeding:** Populate database with comprehensive realistic test data (`02_sample_data.sql`).
- **SQL Views & Queries:** Create operational views and analytical queries (joins, aggregations, window functions).
- **Transaction Management:** Implement atomic transaction engine with pessimistic locking in Python (`library.py`).
- **Dual User Interface:** Build full desktop GUI (Tkinter) and modern Web Application (Flask + Tailwind).
- **Systematic Testing:** Complete unit and integration test suite (`test_library.py`).